Experiment No.: 06

Name of the Experiment: Implementation of Bluetooth and Voice-Controlled Home Automation System using HC-05 and Arduino.

Objectives:

1. To understand the working principle and interfacing of the HC-05 Bluetooth module with Arduino Uno via UART serial communication.
2. To design and implement a home automation system capable of controlling AC appliances wirelessly through a smartphone application.
3. To implement voice command recognition using the "BT Voice Control for Arduino" Android app and transmit text commands to the Arduino via Bluetooth.
4. To demonstrate relay-based switching of multiple AC loads (e.g., lights, fans, TV, AC) based on received Bluetooth/voice commands.
5. To display the real-time status of controlled appliances on a 16×2 LCD module.

Theory:

Home automation refers to the automatic and electronic control of household appliances, security systems, and other household features. The increasing adoption of wireless technologies such as Bluetooth has made it possible to control home appliances remotely from a smartphone, offering users greater convenience, energy efficiency, and safety.

Arduino Uno: The Arduino Uno is an open-source microcontroller board built around the ATmega328P chip. It operates at 5V, has 14 digital I/O pins, 6 analog inputs, and communicates via UART, SPI, and I2C protocols. In this experiment, it serves as the central processing unit that receives serial data from the HC-05 module, processes the received string commands, and drives the relay module accordingly.

Relay Module: A relay is an electrically operated switch that uses a low-power DC signal from the Arduino to control a high-power AC circuit. The 4-channel relay module used in this experiment is active-LOW, meaning a logic LOW (0V) activates the relay and connects the appliance, while a logic HIGH (5V) deactivates it. The relay driver circuit typically consists of an NPN transistor (S8050), a flyback diode (1N4007) to protect against back-EMF, and a status LED.

HC-05 Bluetooth Module: The HC-05 is a commonly used, cost-effective Bluetooth Serial Port Profile (SPP) module that operates at 2.4 GHz with a default communication baud rate of 9600 bps. It supports both master and slave configurations and can be interfaced with Arduino using UART (TX/RX pins). The module has two operating modes: Data Mode (for normal data transmission and reception) and AT Command Mode (for configuration). In this experiment, it operates in Data Mode as a slave, receiving commands wirelessly from the smartphone.

Voice Control Mechanism: The "BT Voice Control for Arduino" Android application uses the smartphone's built-in Google Speech Recognition API to convert spoken words into text strings. These text strings are transmitted to the Arduino through the paired HC-05 Bluetooth module via the UART serial port. The Arduino then compares the received string with predefined commands stored in the program code. If a match is found, the corresponding relay is toggled to turn the appliance ON or OFF. The format of the received command is typically "*command#" where '*' and '#' are start and stop delimiters, respectively.

16×2 LCD Display: A 16×2 Liquid Crystal Display is used to provide real-time visual feedback, showing the name and current ON/OFF status of the appliance being controlled. It is interfaced with the Arduino in 4-bit mode and requires a 10 kΩ potentiometer for contrast adjustment.

Serial Communication (UART): UART (Universal Asynchronous Receiver-Transmitter) is the communication protocol used between the Arduino and HC-05 module. Data is transmitted bit-by-bit asynchronously, with start and stop bits framing each byte. The TX pin of the HC-05 connects to the RX pin of the Arduino, and the RX pin of the HC-05 connects to the TX pin of the Arduino. A voltage divider or level shifter is recommended on the HC-05 RX line since the module operates at 3.3V logic while Arduino outputs 5V.

Required Apparatus:

Table 1.1: Required Apparatus

Sl. No.	Components	Specification	Quantity
1.	Arduino Uno	ATmega328P, 5V, 14 Digital I/O pins, 16 MHz	1
2.	HC-05 Bluetooth Module	Bluetooth 2.0, UART, 9600 baud, 2.4 GHz, 3.3V–5V	1
3.	4-Channel Relay Module	5V DC coil, 10A/250VAC contacts	1
4.	16×2 LCD Display	HD44780 controller, 5V, 16 characters × 2 rows	1
5.	10 kΩ Potentiometer	For LCD contrast adjustment	1
6.	NPN Transistor (S8050)	For relay driver circuit	4
7.	1N4007 Diode	Flyback protection diode	4
8.	LED (Green)	5mm, 2V, 20mA – relay status indicator	4

9.	Resistors	220Ω (LED current limiting), 1kΩ (base resistor)	As required
10.	Jumper Wires	Male-to-male and male-to-female	As required
11.	Breadboard / PCB	For circuit assembly and prototyping	1
12.	Android Smartphone	With "BT Voice Control for Arduino" app installed	1
13.	USB Cable (Type-A to B)	For Arduino programming	1
14.	5V DC Power Supply	For Arduino and circuit	1

Circuit Diagram:

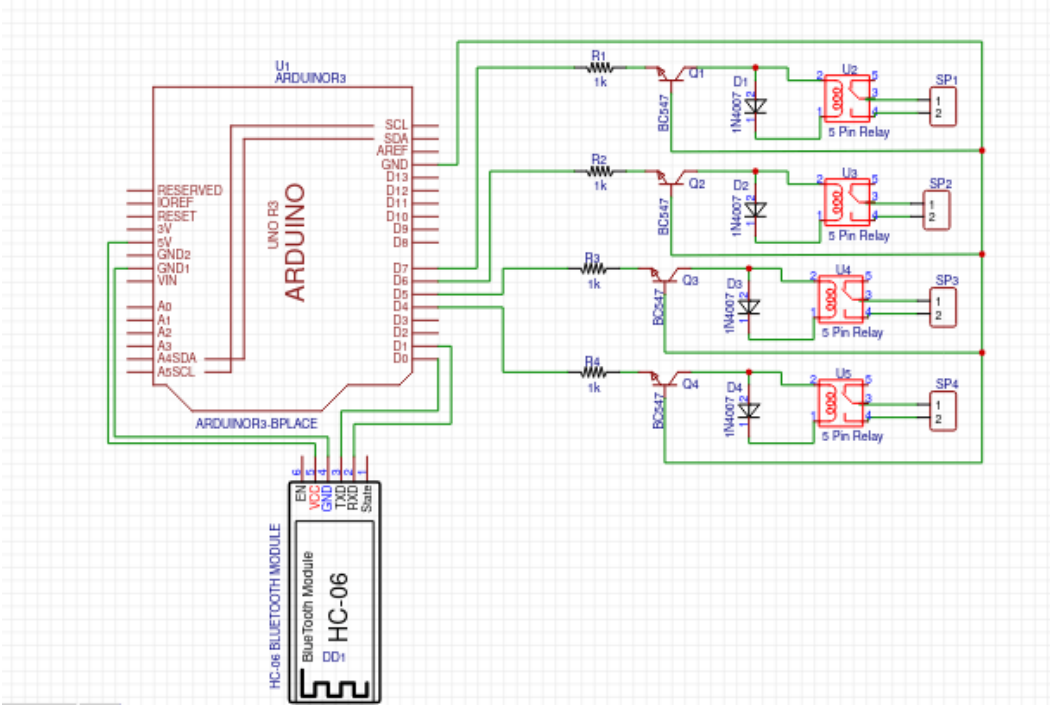


Fig. 6.1: Circuit Diagram of the Bluetooth and Voice-Controlled Home Automation System

Experimental Setup:

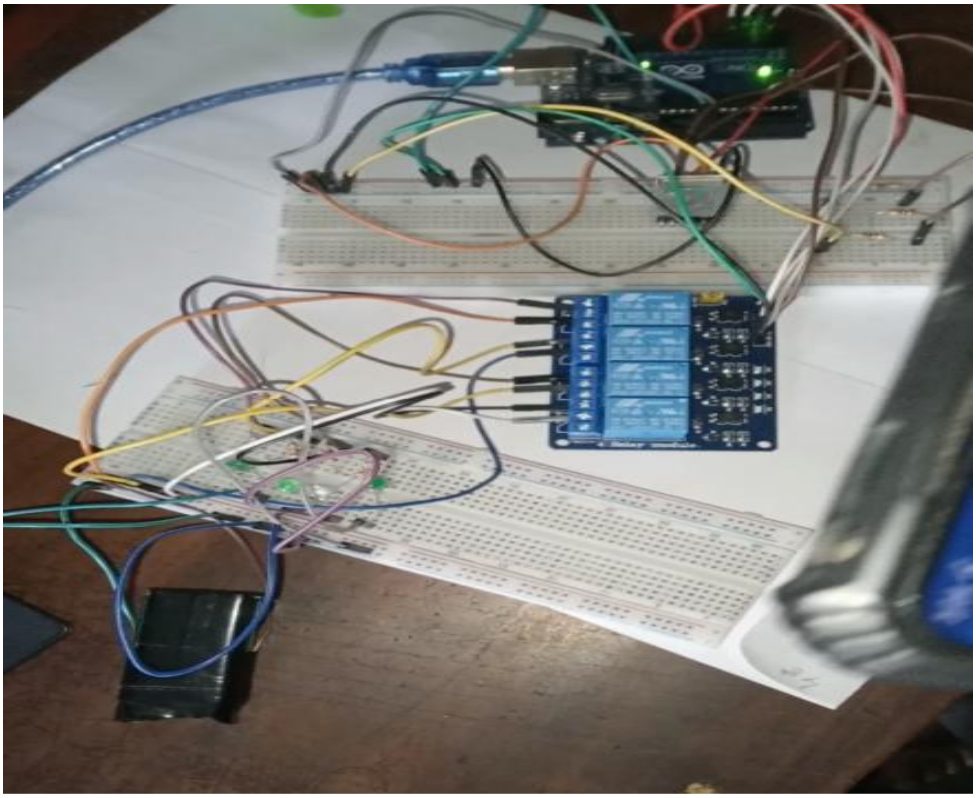


Fig. 6.2: Experimental Setup of the Home Automation System

Arduino Program:

```
#include <SoftwareSerial.h>
#include <LiquidCrystal.h>

SoftwareSerial BT(2, 3);  // RX, TX for HC-05
LiquidCrystal lcd(7, 6, 5, 4, 12, 13);  // RS, EN, D4, D5, D6, D7

const int relay1 = 8;
const int relay2 = 9;
const int relay3 = 10;
const int relay4 = 11;

String voice;

void setup() {
  Serial.begin(9600);
  BT.begin(9600);
  lcd.begin(16, 2);
  pinMode(relay1, OUTPUT);
  pinMode(relay2, OUTPUT);
  pinMode(relay3, OUTPUT);
  pinMode(relay4, OUTPUT);
  digitalWrite(relay1, HIGH);  // Relays are active LOW
  digitalWrite(relay2, HIGH);
```

```
  digitalWrite(relay3, HIGH);
  digitalWrite(relay4, HIGH);
  lcd.print(" Home Automation");
  lcd.setCursor(0,1);
  lcd.print("   System Ready");
  delay(2000);
  lcd.clear();
}

void loop() {
  while (BT.available()) {
    char c = BT.read();
    if (c == '#') break;
    if (c != '*') voice += c;
  }

  if (voice.length() > 0) {
    Serial.println(voice);
    voice.toLowerCase();

    if      (voice == "turn on light")   { digitalWrite(relay1, LOW);
updateLCD("D1 (Light)","ON "); }
    else if (voice == "turn off light")  { digitalWrite(relay1, HIGH);
updateLCD("D1 (Light)","OFF"); }
    else if (voice == "turn on fan")     { digitalWrite(relay2, LOW);
updateLCD("D2 (Fan)  ","ON "); }
    else if (voice == "turn off fan")    { digitalWrite(relay2, HIGH);
updateLCD("D2 (Fan)  ","OFF"); }
    else if (voice == "turn on tv")      { digitalWrite(relay3, LOW);
updateLCD("D3 (TV)   ","ON "); }
    else if (voice == "turn off tv")     { digitalWrite(relay3, HIGH);
updateLCD("D3 (TV)   ","OFF"); }
    else if (voice == "turn on ac")      { digitalWrite(relay4, LOW);
updateLCD("D4 (AC)   ","ON "); }
    else if (voice == "turn off ac")     { digitalWrite(relay4, HIGH);
updateLCD("D4 (AC)   ","OFF"); }
    voice = "";
  }
}

void updateLCD(String device, String state) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(device + " is");
  lcd.setCursor(0, 1);
  lcd.print(state);
}
```

Experimental Procedure:

1. Assemble the circuit on a breadboard according to the circuit diagram, connecting the HC-05 Bluetooth module to Arduino pins D2 (RX) and D3 (TX) via SoftwareSerial, and the relay module inputs to pins D8–D11.
2. Connect the 16×2 LCD to the Arduino in 4-bit mode (RS→D7, EN→D6, D4→D5, D5→D4, D6→D12, D7→D13) with a 10 kΩ potentiometer for contrast control.
3. Disconnect the HC-05 module's TX and RX pins before uploading the Arduino sketch. Upload the program via the Arduino IDE using a USB cable.
4. Reconnect the HC-05 module after successful upload. Power the circuit and observe that the LCD displays the "Home Automation System Ready" message.
5. Install the "BT Voice Control for Arduino" application from the Google Play Store on an Android smartphone.
6. Enable Bluetooth on the smartphone, search for the HC-05 device, and pair it using the default PIN code (1234 or 0000).
7. Open the app, navigate to the "Connect Robot" option, and select the HC-05 Bluetooth module to establish connection. The LED on the HC-05 will change from fast blinking to slow double-blinking upon successful pairing.
8. Press the microphone button in the app and issue voice commands such as "turn on light", "turn off fan", "turn on TV", and "turn off AC". Verify that the corresponding relay activates or deactivates and the LCD updates accordingly.
9. Test all four appliance channels with both ON and OFF commands and record the system's response for each command.
10. Observe the relay status LEDs and note whether the system responds correctly to each voice command with no false triggering or missed activations.

Results and Discussion:

The Bluetooth and voice-controlled home automation system was successfully implemented and tested. After establishing a connection between the Android smartphone and the HC-05 module, voice commands from the “BT Voice Control for Arduino” app were transmitted as text strings to the Arduino via UART.

The Arduino correctly processed each command and controlled the corresponding relays. All appliances—light, fan, TV, and AC—responded accurately, while the 16×2 LCD displayed their ON/OFF status in real time.

The HC-05 provided a reliable range of about 10 meters (line-of-sight). A slight delay (<1 second) was observed due to speech recognition processing. The system operated stably without false triggering.

The active-LOW relay logic functioned correctly, and flyback diodes ensured circuit protection. Relay LEDs offered clear visual status indication.

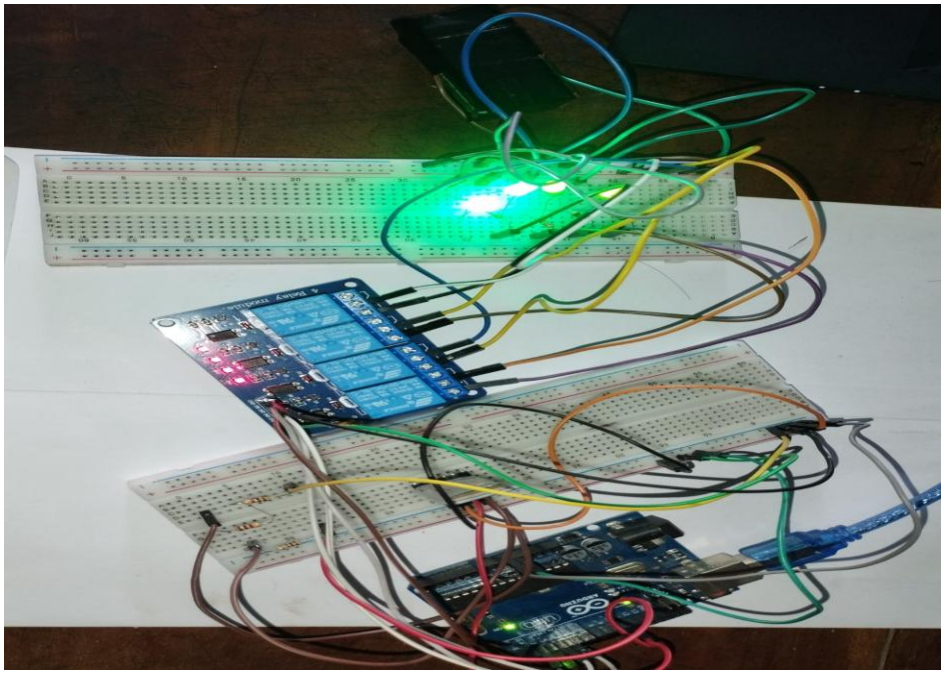


Fig. 6.3: Output results during operation

Conclusion:

This experiment demonstrated a Bluetooth and voice-controlled home automation system using an HC-05 module, Arduino Uno, relay module, and LCD display. The system successfully received voice commands from a smartphone and controlled multiple AC appliances in real time.

It enhanced understanding of UART communication, relay interfacing, and Bluetooth integration with microcontrollers. The use of Software Serial and LCD improved functionality and user feedback. Overall, the system is cost-effective, scalable, and can be extended to Wi-Fi or IoT-based automation.

References:

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5. Circuit Digest, "Bluetooth Controlled Home Automation using Arduino," https://circuitdigest.com